

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2024

Subject: 031001113 WAVE MOTION AND SOUND

Section: 15-18

Date: 9 August 2024

Time: 10:00-12:00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 8 pages (including this page) and they are worth 35 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents, and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

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Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Section 1	Section 2
CLO 1	Do experiment and explain the simple harmonic motion of objects attached to spring ends and simple pendulums. Including calculating various related quantities	1,2,3,4,13,14,15	1
CLO 2	Explain wave phenomena, types of waves, wave parameters, observe and explain various properties of waves, as well as calculate speed, frequency, wavelength, and related quantities.	5,6,7,8,16,17,18	2.1,2.2
CLO 3	Explain and do experiment with sound waves. sound propagation, relation of various quantities of waves, sound level, sound quality, sound resonance in pipes, including observation and explanation of beat formation, standing waves, doppler effect. sound shock wave Including calculating various quantities related to and applying the knowledge of sound to everyday life.	9,10,11,12,19,20, 21,22,23	3,4

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Session 1: Multiple-choice Question (17 questions) (1 mark / question)

Instruction: mark X for a correct choice of each question on the answer sheet provided.

Answer sheet												
	A	B	C	D	E		A	B	C	D	E	
1						13						
2						14						
3						15						
4						16						
5						17						
6						18						
7						19						
8						20						
9						21						
10						22						
11						23						
12												

$$F_s = -kx \quad f = \frac{1}{T} \quad \omega = 2\pi f \quad EPE = \frac{1}{2} kx^2$$

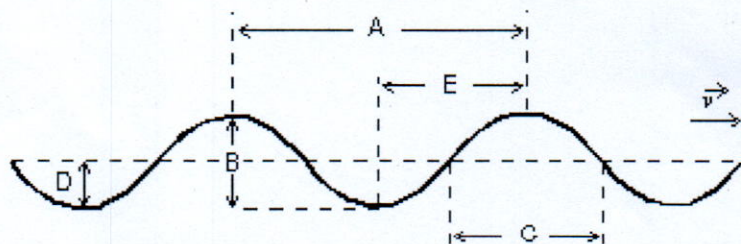
$$x = A \cos \omega t \quad v_x = -A\omega \sin \omega t \quad a_x = -A\omega^2 \cos \omega t$$

$$T = 2\pi \sqrt{\frac{L}{g}} \quad T = 2\pi \sqrt{\frac{m}{k}} \quad v = f\lambda \quad v = \sqrt{\frac{T}{m/L}} \quad f_n = n \left(\frac{v}{2L} \right)$$

$$f_{\text{beat}} = |f_1 - f_2| \quad f_n = n \left(\frac{v}{4L} \right) \quad v = 331 + 0.6T \quad I = \frac{P}{A}$$

$$\beta = 10 \log \left(\frac{I}{I_o} \right) \quad f' = \left(\frac{v + v_o}{v - v_s} \right) f \quad \sin \theta = \frac{v}{v_s}$$

1. Which choice shows the characteristic of restoring force?
 - a. Always directed toward the position of maximum displacement
 - b. Always have the same direction as the motion of the object
 - c. Always constant
 - d. It is zero when the velocity of the object is zero
 - e. It is not constant
2. A body oscillates with simple harmonic motion along the x axis. Its displacement varies with time according to the equation $x = 5.0 \cos(\pi t)$. The magnitude of the acceleration (in m/s^2) of the body at $t = 1.0$ s is approximately
 - a. 3.5
 - b. 49
 - c. 14
 - d. 43
 - e. 4.3
3. The velocity function $v = -0.06 \sin(0.5\pi t)$, where velocity, angular frequency, and time are in the SI unit. What is the highest velocity of this motion?
 - a. 0.015 m/s
 - b. 0.030 m/s
 - c. 0.094 m/s
 - d. 0.148 m/s
 - e. None of the above
4. Which of the following combinations of variables results in the highest frequency for a pendulum?
 - a. Length = L , mass = M , and maximum angular displacement = 3 degrees
 - b. Length = $2L$, mass = $M/2$, and maximum angular displacement = 1 degree
 - c. Length = $1.5L$, mass = $2M$, and maximum angular displacement = 2 degrees
 - d. Length = $\sqrt{2}L$, mass = $\sqrt{2}M$, and maximum angular displacement = $\sqrt{2}$ degrees
 - e. Length = $\sqrt{2}L$, mass = $4M$, and maximum angular displacement = 4 degrees
5. Which choice is not the electromagnetic waves?
 - a. Visible light
 - b. X-ray
 - c. Microwave
 - d. Radio wave
 - e. Sound
6. A sinusoidal wave is traveling toward the right as shown. Which letter correctly labels the amplitude of the wave?



- a. A
- b. B
- c. C
- d. D
- e. E

7. The speed of a sinusoidal wave on a string depends on:
- the frequency of the wave
 - the wavelength of the wave
 - the length of the string
 - the tension in the string
 - the amplitude of the wave
8. A string of mass 50 g and length 64 cm is stretched and fixed at both ends. It is stretched by a force of 5 N and shake it. What is the speed of wave that passing on it?
- 2 m/s
 - 4 m/s
 - 6 m/s
 - 8 m/s
 - 10 m/s
9. What is the distance between the adjacent nodes of standing wave?
- $\frac{\lambda}{8}$
 - $\frac{\lambda}{4}$
 - $\frac{\lambda}{2}$
 - λ
 - 2λ
10. The largest number of beats per second will be heard from which pair of tuning forks?
- 200 and 201 Hz
 - 256 and 260 Hz
 - 534 and 540 Hz
 - 763 and 774 Hz
 - 8420 and 8422 Hz
11. What is the intensity level of sound wave which has intensity of $1 \times 10^{-7} \text{ W/m}^2$?
- 10 dB
 - 30 dB
 - 50 dB
 - 70 dB
 - 90 dB
12. How does the doppler effect occur?
- Source is moving only
 - Observer is moving only
 - Source and observer move toward to each other
 - Source and observer move away to each other
 - All above are correct

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Question 13-23

Read each statement below carefully.

Place an X on the choice A in the answer sheet if you think a statement is TRUE.

Place an X on the choice E in the answer sheet if you think the statement is FALSE.

13. In SHM motion, at the position of the maximum velocity is zero acceleration.
14. The total energy of an object which is moving in SHM is changed depending on time.
15. In simple pendulum, the period of the object is inversely the square root of length of the string.
16. Sound wave is one type of electromagnetic wave.
17. The speed of sound waves in the air decreases when the temperature of the air increases.
18. The speed of sound waves in the air depends only on its frequency.
19. If the tension of the string is increased, the fundamental frequency will increase.
20. Two sound waves are traveling through the same medium. They have the same amplitude, wavelength, and direction of travel. If the path difference is 5λ , the type of interference they exhibit is fully constructive.
21. If the sound level is increased by 10 db the intensity increases by a factor of 10.
22. Beats in sound occur when two waves of slightly different frequency interfere.
23. In Doppler effect, the source being in motion while the observer is at rest. As the source moves toward the observer, the wavelength appears shorter.

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Session 2: Step-by-step explanation is required for this section.

(4 questions) (3 marks/question)

1. A mass $m = 2.0$ kg is attached to a spring that has a spring constant $k = 290$ N/m. The mass is displaced from its equilibrium position and released. Find the frequency of oscillation (in Hz). (3 marks)

2.1 The speed of a 10-kHz sound wave in seawater is approximately 1500 m/s. What is its wavelength in sea water? (2 marks)

2.2 Ocean waves with a wavelength of 120 m are coming in at a rate of 6 waves per minute. What is their speed? (1 marks)

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3. The $64\pi \times 10^{-5}$ W of sound power passed through the surface at the distance of 4 m from source. What is the sound level at this position? (3 marks)

4. Source emits sound waves with the frequency of 1000 Hz. An observer and source are move away to each other with speed of 2 m/s and 6 m/s, respectively. If the speed of sound wave in air is 344 m/s, what is the frequency at the observer? (3 marks)

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